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w5.1- Prof. Harris' Quadcopter Video Demo

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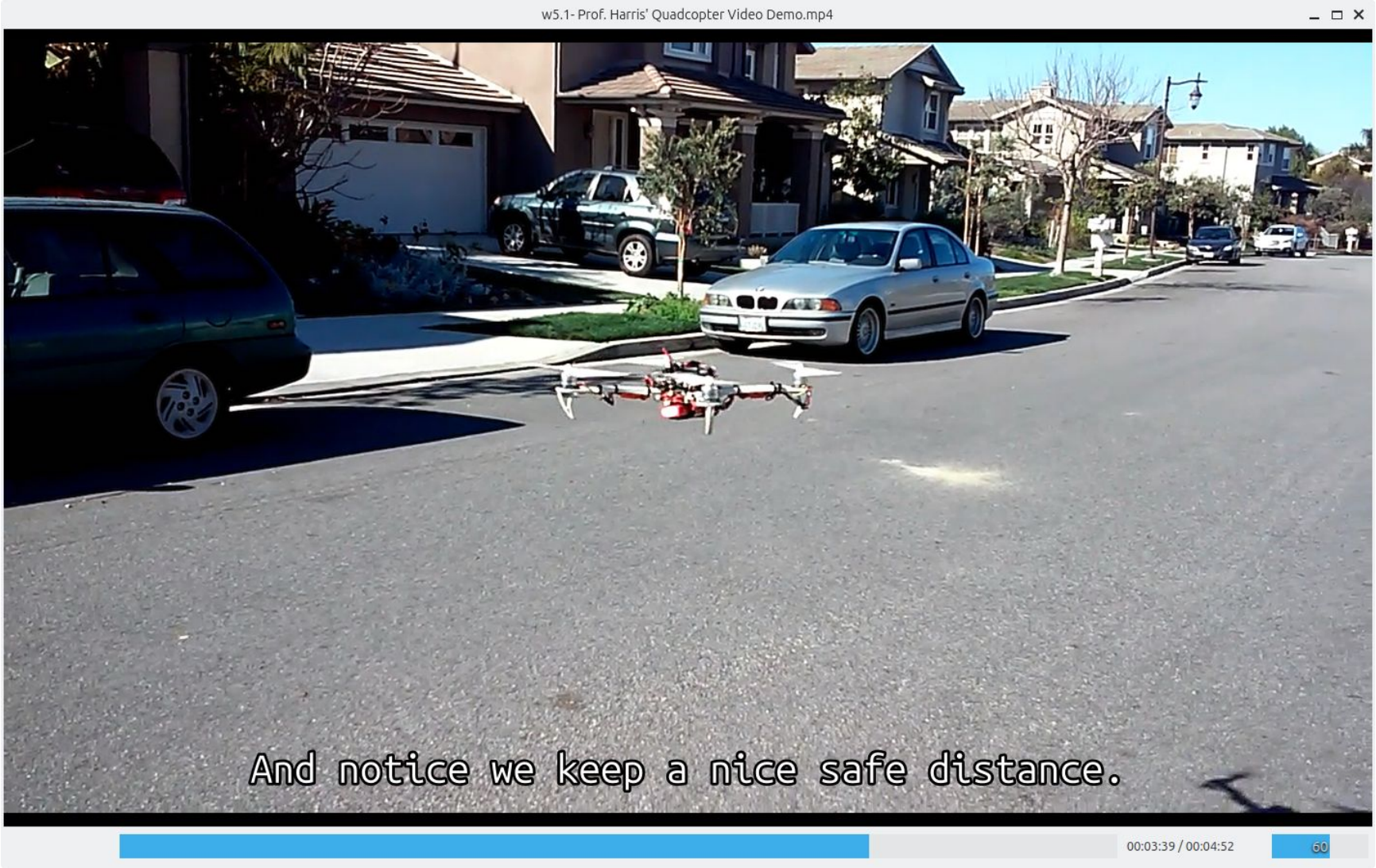

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Summary

## 1. Hardware Components

The quad-copter is built with a few essential "organs" that work together:

- **Propellers & Motors:** Four of each to provide lift.
- **ESC (Electronic Speed Control - 电子调速器):** The "red things" under the motors. They receive signals from the flight controller and tell the motors how fast to spin.
- **APM Flight Controller:** The "center of the action." It processes input from the receiver and coordinates the ESCs.
- **Receiver & Antenna:** Picks up signals from the handheld transmitter.
- **LiPo Battery:** The power source, usually strapped to the bottom.

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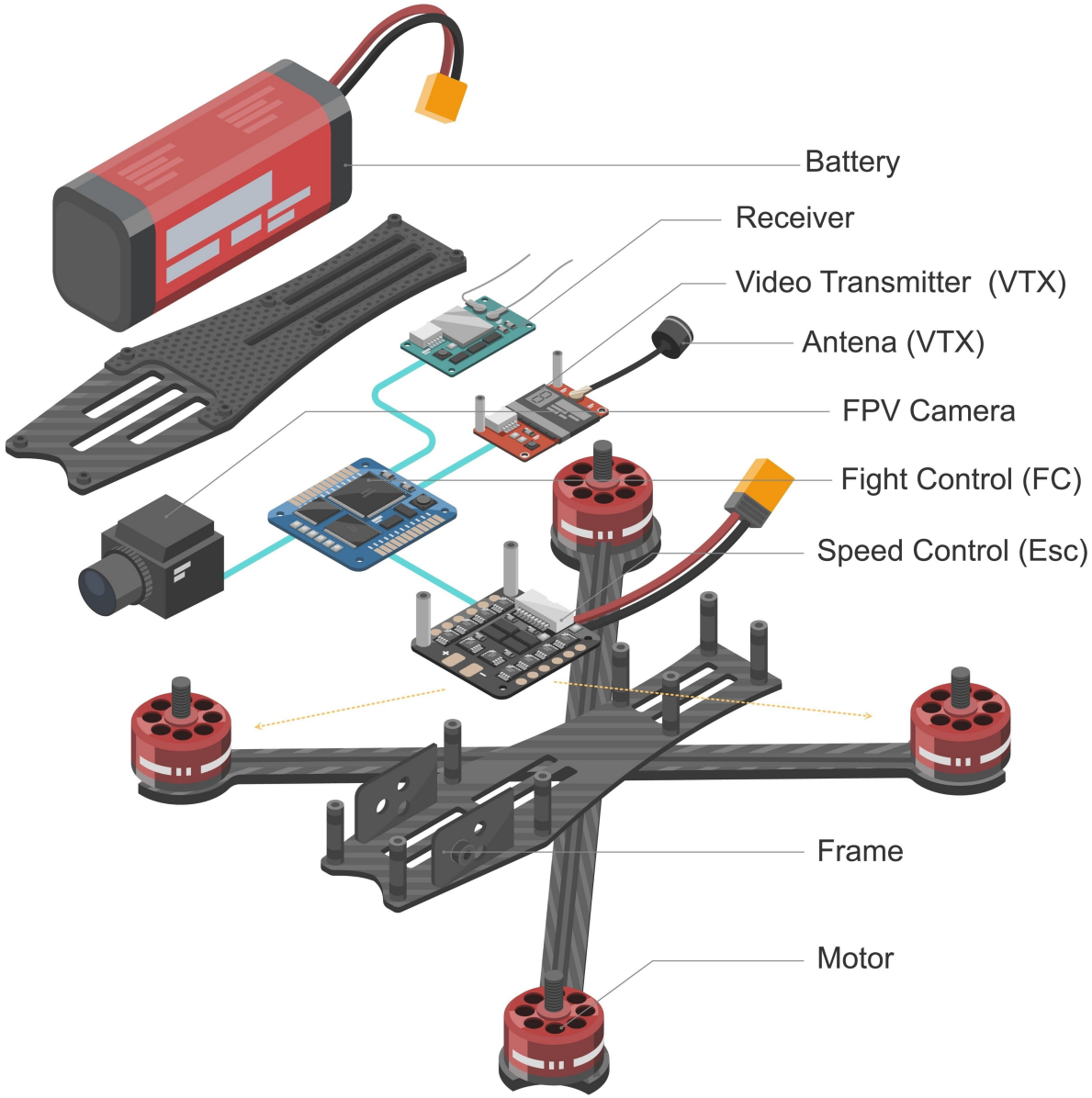

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## 2. Control Mechanics (The Transmitter)

The transmitter uses two joysticks, each controlling two axes of motion. This is standard for drones and something we'll definitely need to master for the rover:

| Control | Action | Effect on Vehicle |
|---|---|---|
| **Throttle** (节气门) | Left Stick (Up/Down) | Controls altitude/speed. |
| **Yaw** (偏航) | Left Stick (Left/Right) | Rotates the vehicle in place. |
| **Pitch** (俯仰) | Right Stick (Up/Down) | Tilts the vehicle forward or backward to move. |
| **Roll** (横滚) | Right Stick (Left/Right) | Tilts the vehicle left or right to move sideways. |

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### 3. Key Observations

- **Frame Rate Illusion:** The propellers might look like they are barely moving on camera. This is just an alias effect—they are actually spinning at high RPMs!
- **Safety First:** The presenter emphasizes keeping a safe distance and only plugging in the battery right before takeoff.
- **Environmental Factors:** Even a small amount of wind (风 - fēng) can significantly impact the stability and "yaw" of the craft.

This quad-copter setup is a great reference for our **Aachen-Sanctuary** projects. For a rover, we'd swap those propellers for wheels, but the ESCs and Flight Controller logic remain very similar.